

1 Event Maintenance

Overview

	Drei kleine textuelle Korrekturen/Änderungen.	
Menu	Data collection → Data correction → Event maintenance	
Transaction code	evli	
Function authorization	evli	

Purpose

You can use the *Event maintenance* to edit and recalculate the posting events provided by the data collection. You can also delete events or create new events.

Integration

The user can use the maintenance function to subsequently change events. The system can then perform a validation check and recalculate the events. The resulting postings can be integrated (selected) in the different evaluations or uploaded to the higher-level ERP system.

The special feature of the event maintenance, compared to the editing functions for postings, is that it includes a recalculation function. The changed data (events) is then used to generate new postings. When data is recalculated, the system performs the same validation checks as for postings of the online data collection. This procedure ensures consistent data.

Requirements

The event maintenance provided with this application requires activation. If the activation is not performed, the following message is shown when the application is called: "The event-related BDE maintenance has been disabled in the basic settings. Modifications cannot be saved." Without activation, the events can only be selected and displayed, but you cannot add new events or edit or delete existing events.

By default, the event maintenance is not active.



Only activate the event maintenance if the relevant requirements or the restrictions listed in the following have been checked.

Restrictions

Some posting functions are pure data collection functions. The collected data cannot be edited or recalculated in the event maintenance. You must check for each function whether the event maintenance can be used. Find a list of these functions in the following.

Data from data collection that cannot be edited

- Comments entered in different posting dialogs on the shop floor terminal cannot be displayed or edited in the event maintenance.
- Serial numbers collected for OPs that require serial numbers cannot be displayed or edited in the event maintenance.
- The system can collect quantities in alternative units. In the event maintenance, you can only edit quantities in primary quantity unit.
- You can post manual activities in the system or calculate them using formulas. You cannot use the event maintenance to show or edit activities.
- Posting with OP reference or confirmation/upload number: In the event maintenance, you require the MES order number for editing.
- Changes of the production lock are logged as events. You can display these events in the event maintenance, but you cannot edit them.
- User fields that were integrated in posting dialogs as part of a customization (e.g. OP logoff) cannot be displayed or edited in the event maintenance.
- Automatic posting processes and their configurations are not used and recalculated in the event maintenance (e.g. finish/interrupt or automatically log off a preceding operation when target quantity is reached).
- Events that are based on resource-related postings (WRM or DNC) cannot be displayed or edited in the event maintenance.
- The "Automatic OP change" option in tab *BDE* of the *HR master data* is only integrated during online processing; it is not integrated during recalculation in the event maintenance.
- The option "OP change with status change" in the status assignment of machines/workplaces is only integrated during online processing; it is not integrated during recalculation in the event maintenance.
- Postings of personnel or machine waiting periods are not displayed in the event maintenance.
- The recalculation in the event maintenance does not trigger escalations.
- Optional data that has additionally been recorded in the "Change machine status" dialog is not displayed in the editing dialog and cannot be changed. This additional data can be the "expected duration", a comment or any additional data for escalation management.

Merged operations

An assignment of a new OP to a merged operation is NOT supported in the event maintenance. Only data for already posted merged operations can be modified.

If data is recorded for a merged operation using the "Merged operation per machine" function, this data CANNOT be edited in the event maintenance.

A recalculation is no longer possible for merged operations that were generated on the MOC if the operations contained in the MOP were changed.

PZE controls ADE

The "PZE controls ADE" setup setting can lead to scenarios in the event maintenance where the personnel logon event is displayed before the actual order logon, because the logon is then rounded to the PZE time.

If inconsistent postings exist in the event maintenance – because a personnel logon is displayed before the actual order logon – then the postings must be corrected. The personnel logon time must be changed to the time of the order logon.

Interface to SAP

The event maintenance cannot be used in combination with the PP-PDC interface, because the standard SAP system does not accept cancellations from external systems.

MDE shift change events

If these events are changed, it is possible that shifts are not calculated completely or that gaps result. For this reason, generally do not edit shift change events.

A recalculation of the entire shift is not guaranteed here because the relevant events that identify the times of start and end of shift might no longer be available at the relevant point in time.

Shift change events are events triggering status changes "M_MST"

end of shift → dialog A_AUN

beginning of shift → dialog A_AAN

Shift end events (event M_MST with dialog A_AUN) may only include the status 20000 "no shift".

Editing of finished operations

Operations that have already been finished can be edited in the event maintenance dialog. Note: The status of a finished operation does not change. The status is also not changed if an OP logoff is deleted or if the OP logoff is changed into an interruption.

If you want to reactivate an operation that has already been finished, you must always use the function "Reactivate operation".

Material and Production Logistics (MPL)

Some operations require batch management. The following restrictions apply for these operations in the event maintenance:

- Only MPL machines with batch processing can be edited. Machines of the types roll cutting, parallel output batches, throughput batch mode, etc. are not supported.
- The event maintenance supports input batch postings for an operation to a limited extent only. Postings that are not included in the OP posting are not integrated. Consequently, consumption postings for input batches are not integrated.
- It is not allowed to change output batch numbers or to delete events CA_AN/CA_AB.
- Using the event maintenance, you cannot display or change batch attributes.
- The existing batch tracing is not changed subsequently by the maintenance function.

Event modified by

You can store a user ID for the event. For technical reasons, the user ID has a maximum of 9 digits (column "Modified by"). This means: If you enter a user ID with 10 digits, the last digit is cut when the user is transferred to the "Modified by" column in the event maintenance.

Editing of the partitioning

If a machine partitioning event (M_TLG) is edited, only the shifts included in the recalculation are integrated.

If an order is produced in several shifts (automatic re-login), **all** events must be recalculated that are affected by this change of partitioning event.

Respect the following process for an event that includes a change of the partitioning:

- If partitioning is changed, the entire period for which this modification applies needs to be selected, i.e. until the operation is interrupted or finished manually.
- When changes are made, the relevant partitioning event must be changed
AND
if the operation was automatically produced during several shifts, the subsequent manual interrupt or finish posting of the order must also be changed.
If you manually interrupt or finish the order, it is enough to save the data without change.

- When data is then recalculated, all data between the change of partitioning and the manual interrupt or finish posting is automatically used for this recalculation.

Note:

The events that must be recalculated are not selected automatically, because the change of partitioning can affect several shifts; also shifts that are no longer included in the selection period.

Selection criteria

The application provides the following selection criteria:

Workplace/machine

All data relating to the selected workplace is shown.

MES order number

The result list shows all events of workplaces where the order was logged on in the selected period of time. You can also use wildcards.

Person ... to ...

The result list shows all events of the workplaces where the person (persons) was (were) logged on in the selected period of time. You cannot use wildcards.

Date ... to ...

All events that are included in the selected period of time are displayed.



Note: This application only selects and edits data that is included in the online data area.

Parallel staff logins

The option "Parallel staff logins" selects the data of all workplaces where a person was logged on. Set this option if you want to edit data of workplaces where persons are logged on with multiple machine operation.

For more information, see [here](#).

Refresh data

Use this option to control if the new data is displayed in the event maintenance after recalculation (with large amounts of data, this can have a negative effect on performance). Note: If an error occurred during recalculation, the data freezes in the display.



If more than 1000 events are selected according to the data selection, the following error message is displayed: "The requested amount of data is too large. Please restrict the data using selection criteria."

When data is requested, all selection fields are disabled except the checkbox "Refresh data".

The fields are only enabled again when the button "Discard" is clicked.

If you request data and this data is currently locked by another user, then you can specify if the data is displayed. Here, the data is only displayed and cannot be edited.

Toolbar

You can call the following functions from the toolbar. Note the [Requirements](#) for editing events:



Edit an existing event

Function authorization: evli.edit

The relevant dialog for the event selected is opened.



Delete an existing event

Function authorization: evli.delete

The selected event is deleted.



Recalculate

After the confirmation prompt, a validation check is performed for all added, edited or deleted events; the postings resulting from the events are regenerated. When the recalculation is completed, a message is issued.

If an error occurs during recalculation (e.g. because of logically invalid values), an error message is displayed.



Discard

After confirmation, all added, edited or deleted events are discarded and the list is cleared.



Order information

Function authorization: orin

Calls the application [Order information](#).



Workplaces/machines

Function authorization: wpov

Calls the application [Workplaces/machines](#).

Tab Create event

Function authorization: evli.create

This tab includes the buttons  to create events. Different categories are available:

Operation category:

- Log OP on (A_AN)
- Log OP off (A_AB)
- Partial confirmation/posting of part quantity (A_TR)
- Quantity upload (A_MR)
- Interrupt OP (A_UN)
- Finish OP (A_BE)

Person category:

- Log person on (P_AN)
- Log person off (P_AB)

Machine/workplace category:

- Change status (M_MST)
- Change partitioning (M_TLG),
- Automatic counter (M_CTR_AUTO),
- Automatic quantity (M_AUTO)

Material category:

- Log output batch on (CA_AN)
- Log output batch off (CA_AB)
- Log input batch on (CE_AN)
- Log input batch off (CE_AB)

Other category:

- Activate OP (NC_AN)
- Deactivate OP (NC_AB)

Field descriptions

Class

Internal event classification:

P = Personal data

M = Workplace data

A = Order data

C = Batch data

Event

The possible events and their colors are listed [here](#).

Dialog

This field shows the dialog that triggers the event. Some dialogs have the same name as the event (e.g. A_AN, A_UN, ...), but a different dialog can also trigger the event.

Date, time

Date and time when the event was posted.

Workplace/machine

The event was posted for the workplace specified.

MES order number

The event was posted for the specified combined order/operation number. This field is only filled in if the event has an order reference.

Person

The person who triggered the event or for whom the event was triggered. This field is only filled in if the event was triggered by a person or has a reference to a person.

Status

The workplace/ machine status that applied at the time of the event; this field contains only context-related information.

PZE

Reference to clock-in or clock-out

Primary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Secondary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Tertiary quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Base quantity

Yield, yield reason, scrap, scrap reason, rework quantity, rework quantity reason, open quantity, open quantity reason

Modified by

Last editor of the event

Date, time

Date and time when the data record was last edited.

Editable

J = Event can be edited

N = Event cannot be edited, e.g. is the master of an MOP or is locked by a change in the log record

Reference

Unique ID of the data record

Priority

Priority specification for events of the same time (1 = highest priority)

Sorting

Internal use

Status 1

Event M_MST: machine status/ interruption reason

Event M_TLG: Partitioning

Event M_SZY: Target cycle

Status 2

Internal use

Status 3

Internal use

Attribute 1

Events C_GEN, C_UMB, CE_AN, CE_AB, CA_AN, CA_AB: target location/material buffer

Attribute 2

Event P_AN: Wage/ premium indicator

Events CE_AB, C_GEN: info on batch

Otherwise: Internal use

Attribute 3

Event P_AN: Operator position

Events CA_AB, C_GEN: transport unit

Event CE_AN: BOM item

Event CE_AB: Batch status

Attribute 4

Internal use

Attribute 5

Event CE_AB: Current batch

Counter 1, Type 1, Reason 1

Internal use

Counter 2, Type 2, Reason 2

Internal use

Counter 3, Type 3, Reason 3

Internal use

Counter 4-10, Type 4-10, Reason 4-10

Internal use

Partitioning

Partitioning. Is currently only displayed for the event M_TLG.

Displayed events

The following events are displayed:

Event	Designation	Type
A_AN	Log operation on	Order-related event
A_TR	Partial confirmation / posting of part quantity	Order-related event
A_UN	Interrupt operation	Order-related event
A_AB	Log operation off	Order-related event
A_BE	Finish operation	Order-related event
A_MR	Quantity upload You can use the quantity upload to upload quantities for operations, which are not logged on at the moment. This way, you can correct a quantity of an operation without having to log the operation on and off.  If an operation is logged on, you may not use this event. Use instead the event A_TR.	Order-related event
P_AB	Log person off	Person-related event
P_AN	Log on person	Person-related event
P_VAN	Person advance logon	Person-related event

Event	Designation	Type
		Cannot be edited. The event is not used during recalculation.
M_MST	Change workplace/machine status	Machine-related event
M_AUTO	Automatic quantity posting from the terminal	Machine-related event
M_CTR_AUTO	Automatic counter posting from terminal	Machine-related event
M_TLG	Change of partitioning	Machine-related event
M_SZY	Change of target cycle	Machine-related event Cannot be edited. If the target cycle is changed, the change is not used during recalculation.
M_PSPERRE	Production lock	Machine-related event Cannot be edited. Production lock events are not used during recalculation.
CA_AB	Log output batch off (MPL)	Batch-related event
CA_AN	Log output batch on (MPL)	Batch-related event
CE_AB	Log input batch off (MPL)	Batch-related event
CE_AN	Log input batch on (MPL)	Batch-related event
NC_AB	Deactivate OP (ADE-BEA)	Other event
NC_AN	Activate OP (ADE-BEA)	Other event

The colors of the listed events are the following:

Color	Meaning
blue	Order-related events
green	Person-related events
red	Machine-related events
brown	Batch-related events

Color	Meaning
black	Locked events
purple	If the order type option "Change after upload" is set to "Allow no changes" or "Allow modification, no upload", then all the events already uploaded to the ERP system are displayed in purple.

The column "Dialog" shows the dialog that triggered the event. Some dialogs have the same name as the event (e.g. A_AN, A_UN, ...). The following other dialogs can also trigger the events mentioned above:

Dialog	Meaning
A_P_AN	Log order + person on in one (A_AN + P_AN)
P_AAB	Log all staff off (1..n P_AB)
A_AUN	OP is automatically interrupted with shift change
A_AAN	OP is automatically logged on again with shift change
SA_AN	Log on merged operation
SA_TR	Partial confirmation for merged operation
SA_AB SA_ABME	Log merged operation off
P_KOM	PZE Clock-in
P_GEH	PZE Clock-out
CA_WL	Output batch change

Display of automatically recorded counter values:

If quantities are automatically recorded for a workplace using the Windows terminal software, then these unevaluated counter quantities are collected and posted as event M_CTR_AUTO. The counter values of the counter events M_CTR_AUTO are not displayed in the columns for yield, scrap, etc., but in separate columns of the category "counter".

The primary quantities resulting from the counter values (evaluated) are logged and displayed as information in the columns for yield, scrap, etc.



The resulting (evaluated) quantities are not logged in other quantity units. This means: The base, secondary and tertiary quantities of the counter events are always 0.

No reason is added to the logging of the resulting (evaluated) quantities.

Recalculation of changed data



When you have completed the editing, click button  and confirm the prompt to start the recalculation function. The recalculation can take some time depending on the extent of changes made.

The following steps are performed during recalculation:

1. Validation check of the modified data

The system checks if all events involved are fully processed (same check as with online processing). If the validation check fails for one of the events, then the recalculation is rejected and the changes are not accepted in the system. If a validation error occurs, then an error message is displayed including information on the relevant event.

2. Canceling existing results

If all validation checks were completed successfully, then the still available, current results are canceled. If postings have already been uploaded to the ERP system (upload identifier "J"), then the relevant cancelation records are generated; this is the same processing when postings are manually changed using the editing function for postings. The quantities and times contained in the order-related posting are also canceled for the operation status.

3. Calculating new results

After a successful validation check and cancellation, the modified events are reevaluated and a new posting is generated. The quantities and durations calculated for the order-related posting are also posted to the operation status.



Note: If the configuration has been changed in the meantime, it is possible that you cannot change events any more that have already been recorded or the changed configuration can lead to other results.

Example: If two operations were logged on to a machine at the same time, and in the meantime the option *Parallel logon of OP* was deactivated, then the recalculation of one operation leads to a validation error. Recalculation is no longer possible.

If the recalculation could be completed successfully, a success message is shown.

Locking concept in the event maintenance

The selected data is locked for editing. The selected machine and the selected period of time specify the data locked.

All locks of the event maintenance are displayed in *System administration > Locked data records* and can only be deleted by a user with the relevant function authorization.

If a lock is deleted, the recalculation of the currently displayed data is no longer possible on this console. The user must discard the scenario and request data again.

Option "Parallel staff logins" in the event maintenance

If the employees log on to multiple machines, you must activate this option in the event maintenance selection area. The personnel times are then allocated according to the order postings.

All machines are then displayed where the displayed persons were logged on during the evaluation period.

If a nesting of the data is available, this nesting might be too complex and the data cannot be recalculated because the start events are no longer included in the period of time selected.

Example:

Person 1 is logged on to workplaces 100 and 200

Person 2 is logged on to workplace 200 at a different, but overlapping time

Person 3 is logged on to workplaces 200 and 300 at a different, but overlapping time

When data is recalculated, all events of an order are integrated. With nested order events, a great number of order events must be recalculated. Here, a constellation is possible that does no longer allow a recalculation.

Note: You cannot use the event maintenance with such complex nested personnel postings.

Option "Optimized parallel staff logins" in the event maintenance

If this option is selected, the following optimized processing is performed with parallel staff logins/personnel postings:

- Only the data specified via selection is displayed or requested (machine, order/OP, person)
- Only if a recalculation is performed, the machines are selected that must be recalculated because of parallel staff logins.
The system uses the changed data to identify the relevant machines.
- If an error occurs during recalculation, the additional machines are also displayed and the user can correct the data.

Waiting period processing

Waiting period processing is an optional processing that controls the system behavior when personnel postings are collected. You activate the processing in the basic settings.

If the waiting period is exceeded, a separate waiting period posting is generated (personnel posting to the defined waiting period operation). If the waiting period is not exceeded, the personnel posting is backdated (and also the OP posting, if required).

The posting times of the events remain unchanged during waiting period processing, i.e. the posting times are edited in the event maintenance.

If the events are recalculated, the changed postings are used to identify whether the times must be backdated according to the waiting period processing.

Changes after upload

As part of the customization, you specify for each order type whether changes made in the event maintenance are uploaded to the ERP system. The following options are available:

- Allow modification and upload
- Allow modification, no upload - You can change data, but the data is not uploaded
- Do not allow modification - data cannot be changed.

The validation check is performed for the workplace: If an order at a workplace cannot be edited according to the order type configuration (purple font color), then all events are locked that are older than the last upload date/time of this order. The lock is performed for the workplace, i.e. also orders that could be edited, are locked.